



AI-Assisted Structural Analysis of a Building Floor Plan

Homework Assignment 1 - Floor Plan 1

Course	Computer (& AI) Methods for Structural Engineering (DISC 5203)
Student	Shuhan He
Student ID	21345221
Assigned plan	Floor Plan 1 (student-ID final digit 1)
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Net loaded floor area: 180.13 m2 | One-floor gravity load: 1576.1 kN | Five-storey base load: 7880.6 kN

Scope

This submission extracts the floor outline and vertical supports, assigns tributary area to each column and wall, calculates dead, live and total axial loads, verifies global equilibrium, and documents a reproducible AI-agent workflow.

1. Input and engineering interpretation

The assignment PDF and the supplied FloorPlan1 image were inspected visually. Printed dimensions govern the metric geometry; pixel dimensions are used only for visual review. The local datum is the lower-left principal wall-centreline intersection, with positive x to the right and positive y upward.

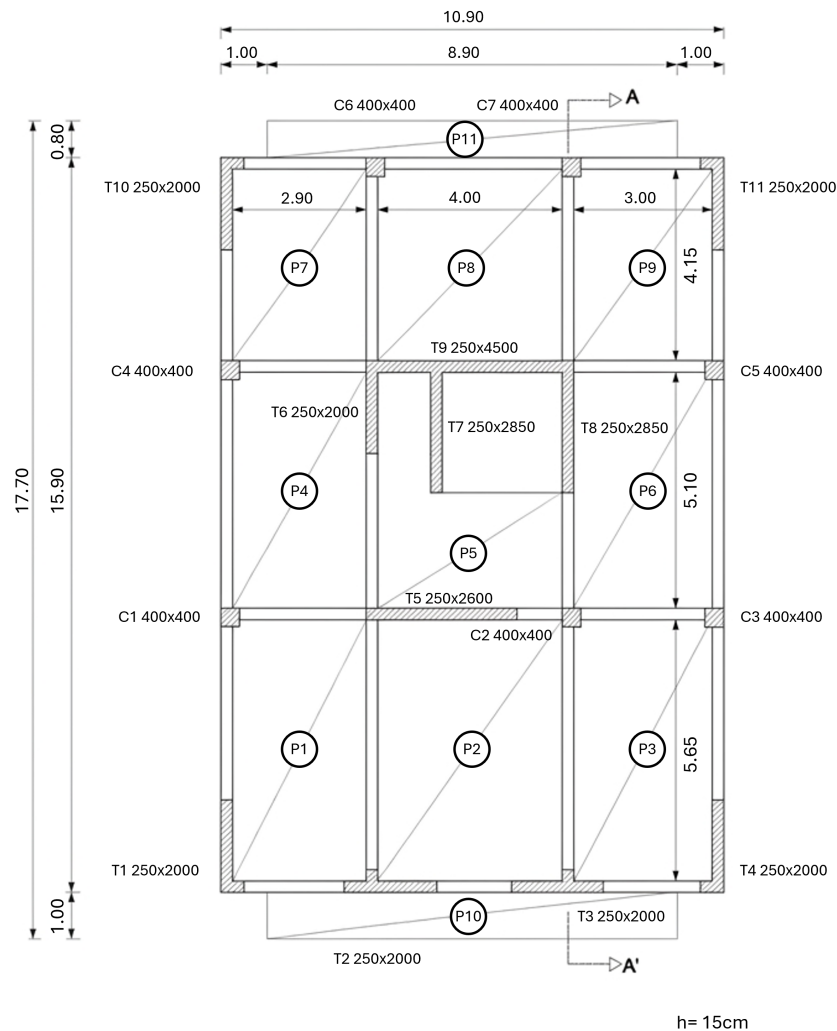


Figure 1. Assigned source image, Floor Plan 1.

The source is a raster drawing rather than CAD. Explicit dimensions were used directly. Where endpoints or core opening limits are not fully dimensioned, coordinates were inferred from adjacent dimension chains and recorded in the JSON configuration for audit and revision.

2. Extracted geometry and vertical elements

The modeled slab has a gross area of 189.33 m². Two core/stair openings totaling 9.20 m² are removed, leaving a net floor area of 180.13 m². Seven columns and eleven finite wall segments are represented.

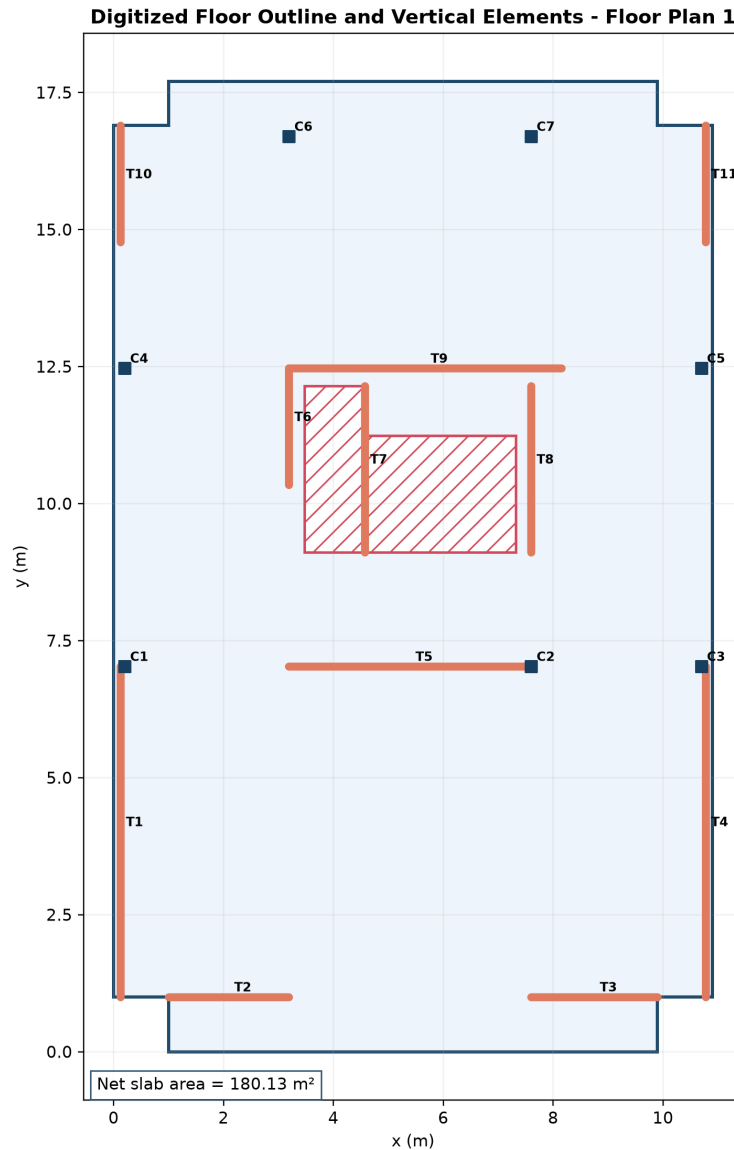


Figure 2. Digitized floor outline, excluded openings, column centers, and wall centrelines.

3. Tributary-area method

The net slab is divided into 0.02 m square cells. Each cell centre is assigned to the closest vertical element. Columns are point supports. Walls are finite line segments, so the distance is measured to the nearest point on each wall axis. This is a generalized Voronoi partition and reproduces mid-span boundaries and 45-degree corner transitions without manually drawing every bisector.

For point p and wall segment a - b : $t = \text{clamp}(((p-a) \cdot (b-a)) / \|b-a\|^2, 0, 1)$; $d = \|p - (a + t(b-a))\|$.

The counted raster area is adjusted by one global factor equal to exact polygon area divided by raster area. The boundary-cell error before adjustment is 0.0270%, and the final partition closes exactly to the polygon area.

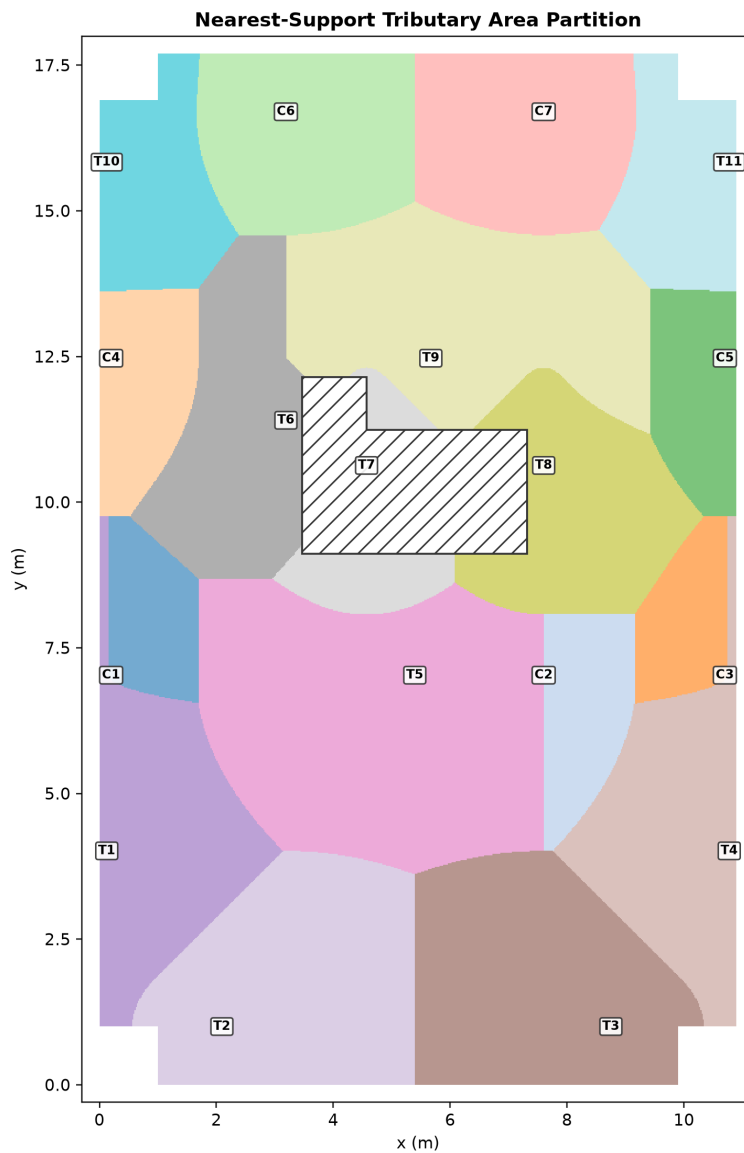


Figure 3. Nearest-support tributary-area partition.

4. Gravity-load calculation

Only the loads supplied by the assignment are included. Concrete slab self-weight is the dead load. Beam, wall, column, finish, facade and partition self-weights are excluded because the necessary weights are not provided.

Quantity	Value	Use
Slab thickness, h	0.150 m	Assignment input
Concrete unit weight, γ	25.0 kN/m ³	Assumed normal reinforced concrete
Dead area load, q_D	3.75 kN/m ²	$q_D = \gamma \times h$
Live area load, q_L	5.00 kN/m ²	Assignment input
Total area load, q	8.75 kN/m ²	$q = q_D + q_L$
Identical storeys	5	Assignment input

For element i : $D_i = q_D A_i$; $L_i = q_L A_i$; $N_{i,1} = D_i + L_i$; $N_{i,\text{base}} = 5 N_{i,1}$. Results are unfactored service loads. The 3 m storey height does not change the slab gravity load and is retained as contextual geometry.

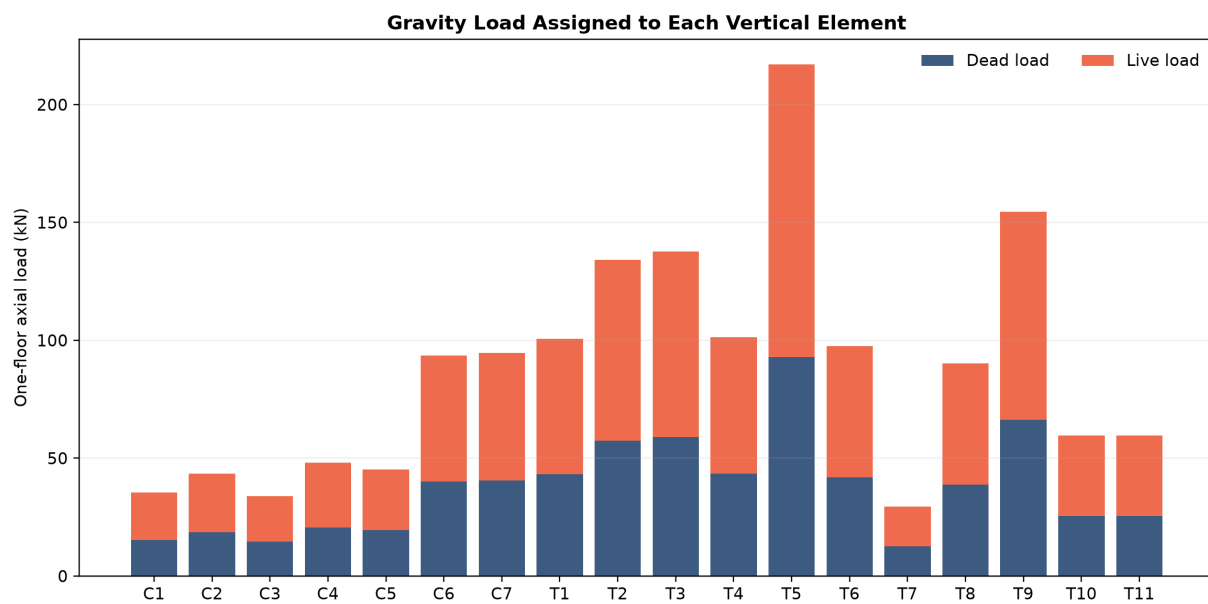


Figure 4. One-floor dead and live axial-load components.

5. Element results

Locations are column centers or wall-segment midpoints. Base total is the cumulative load from five identical floors.

ID	Type	x	y	Size	Area	Dead	Live	1-floor	5-storey
C1	Column	0.20	7.03	400x400	4.05	15.2	20.3	35.4	177.2
C2	Column	7.60	7.03	400x400	4.98	18.7	24.9	43.5	217.7
C3	Column	10.70	7.03	400x400	3.87	14.5	19.3	33.8	169.2
C4	Column	0.20	12.47	400x400	5.49	20.6	27.4	48.0	240.0
C5	Column	10.70	12.47	400x400	5.17	19.4	25.8	45.2	226.1
C6	Column	3.19	16.70	400x400	10.69	40.1	53.5	93.6	467.8
C7	Column	7.60	16.70	400x400	10.82	40.6	54.1	94.7	473.5
T1	Wall	0.12	4.01	250x2000	11.50	43.1	57.5	100.6	503.2
T2	Wall	2.10	1.00	250x2000	15.32	57.4	76.6	134.0	670.2
T3	Wall	8.75	1.00	250x2000	15.73	59.0	78.6	137.6	688.0
T4	Wall	10.78	4.01	250x2000	11.59	43.5	58.0	101.4	507.1
T5	Wall	5.39	7.03	250x2600	24.80	93.0	124.0	217.0	1084.9
T6	Wall	3.19	11.40	250x2000	11.16	41.8	55.8	97.6	488.2
T7	Wall	4.57	10.63	250x2850	3.36	12.6	16.8	29.4	147.1
T8	Wall	7.60	10.63	250x2850	10.31	38.7	51.6	90.3	451.3
T9	Wall	5.67	12.47	250x4500	17.66	66.2	88.3	154.5	772.5
T10	Wall	0.12	15.84	250x2000	6.82	25.6	34.1	59.7	298.3
T11	Wall	10.78	15.84	250x2000	6.82	25.6	34.1	59.7	298.3

Units: coordinates in m; size in mm; area in m²; loads in kN.

6. Building-level accumulation

Level	Floors supported	Dead load (kN)	Live load (kN)	Total load (kN)
Roof/F5	1	675.482	900.642	1576.124
F4	2	1350.964	1801.285	3152.249
F3	3	2026.446	2701.927	4728.373
F2	4	2701.927	3602.570	6304.497
F1/Base	5	3377.409	4503.212	7880.622

7. Verification

Area closure: $\sum(A_i) = 180.128500 \text{ m}^2$, equal to the independent polygon area. Global one-floor dead load = 675.482 kN and live load = 900.643 kN; both equal the corresponding element sums within floating-point precision.

Every in-slab cell is assigned exactly once and all 18 vertical elements receive positive tributary area. Dimensional analysis gives $(\text{m}^2)(\text{kN}/\text{m}^2) = \text{kN}$. The cumulative base result equals exactly five times the one-floor result because the plan and loads repeat on all five storeys.

8. Limitations

The raster source requires interpretation of some wall endpoints and opening edges. The nearest-distance partition is a transparent geometric idealization; a plate analysis could redistribute load according to stiffness and span direction. Non-slab dead loads are omitted because they are unspecified. These limits are explicit rather than hidden, and the configuration can be revised without changing the solver.

9. Reproducible AI-agent workflow

The agent read the source PDF and images, created a reviewable metric geometry model, implemented a deterministic partition and load calculation, generated plots and tables, and ran automated conservation checks. The repository contains the original inputs, geometry JSON, Python scripts, guidance files, a pre-commit verification hook, intermediate outputs, final results, and this report.

Reproduction command sequence

```
python -m pip install -r requirements.txt
python scripts/analyze_floor_plan.py
python scripts/build_report.py
python scripts/render_report_pdf.py
python scripts/verify_outputs.py
```

10. Conclusion

Floor Plan 1 has a modeled net loaded area of 180.13 m². Under $q_D = 3.75 \text{ kN}/\text{m}^2$ and $q_L = 5.00 \text{ kN}/\text{m}^2$, it transfers 1576.1 kN per typical floor and 7880.6 kN at the base of the five-storey stack. The element schedule provides the corresponding tributary area and axial load for each identified column and wall.